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Course Name:5G 6G Communication Systems

Course Code:ECA5601

Ex:8 Steering Angle Variation in MIMO Beamforming Using MATLAB

Aim:

To analyze the effect of steering angle variation in a MIMO beamforming system by studying steering angle, array gain, and beam direction error using MATLAB.

Objective:

1. To analyze the steering angle of a MIMO beamforming system by directing the beam towards different target directions using MATLAB.
- 2.To compare the array gain achieved for different steering angles in a MIMO beamforming system using MATLAB.
- 3.To evaluate the beam direction error between the desired steering angle and the actual beam direction using MATLAB.

Software Required:

- MATLAB

Theory

MIMO (Multiple-Input Multiple-Output) beamforming enhances wireless communication by employing multiple antenna elements to direct transmitted energy toward a desired user. By adjusting the steering angle through appropriate phase shifts, the beam can be focused in different directions, improving signal strength and reducing interference. Important performance parameters include steering angle, array gain, and beam direction error, which collectively determine the effectiveness of beam steering in modern wireless systems such as 5G NR.

Objective 1: Matlab Code and Output

```
clc;
```

```
clear;
```

```
close all;
```

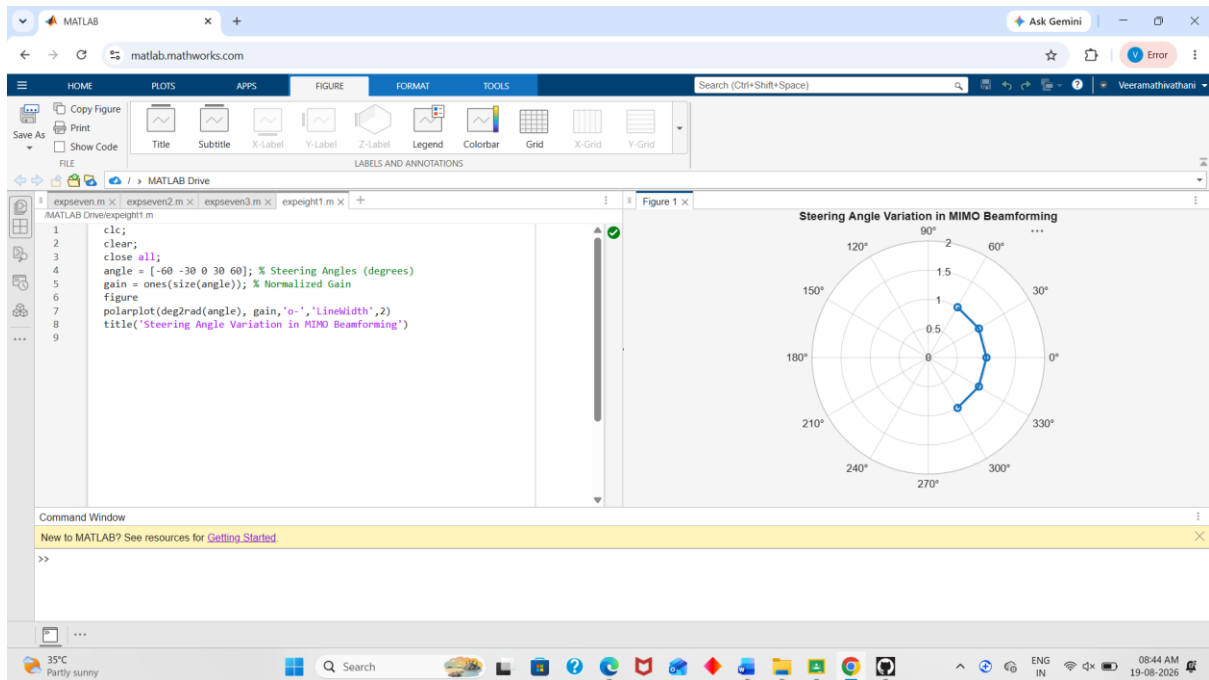
```
angle = [-60 -30 0 30 60]; % Steering Angles (degrees)
```

```
gain = ones(size(angle)); % Normalized Gain

figure

polarplot(deg2rad(angle), gain,'o-','LineWidth',2)

title('Steering Angle Variation in MIMO Beamforming')
```



Objective 2: Matlab Code and Output

```
clc;

clear;

close all;

angle = [-60 -30 0 30 60]; % Steering Angles (degrees)

gain = [9 12 15 12 9]; % Array Gain (dB)

figure

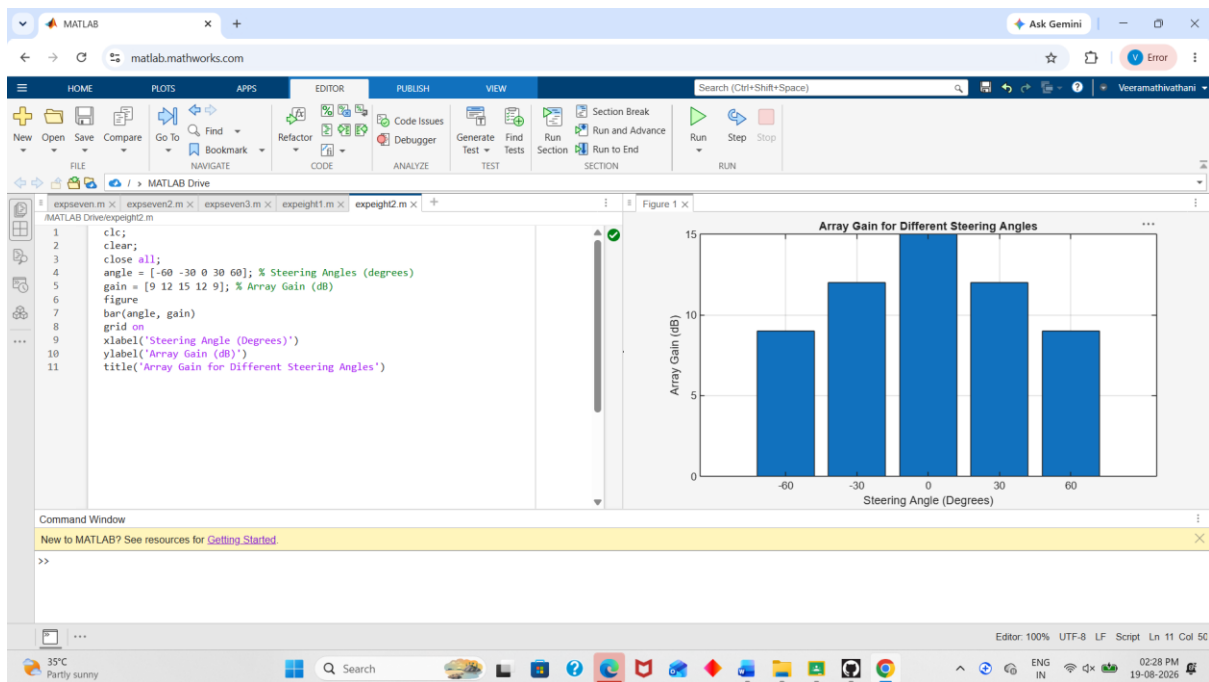
bar(angle, gain)

grid on

xlabel('Steering Angle (Degrees)')

ylabel('Array Gain (dB)')
```

title('Array Gain for Different Steering Angles')



Objective 3: Matlab Code and Output

clc;

clear;

close all;

desired = [-60 -30 0 30 60]; % Desired Steering Angle (degrees)

actual = [-58 -32 1 29 62]; % Actual Beam Direction (degrees)

error = abs(desired - actual); % Beam Direction Error

figure

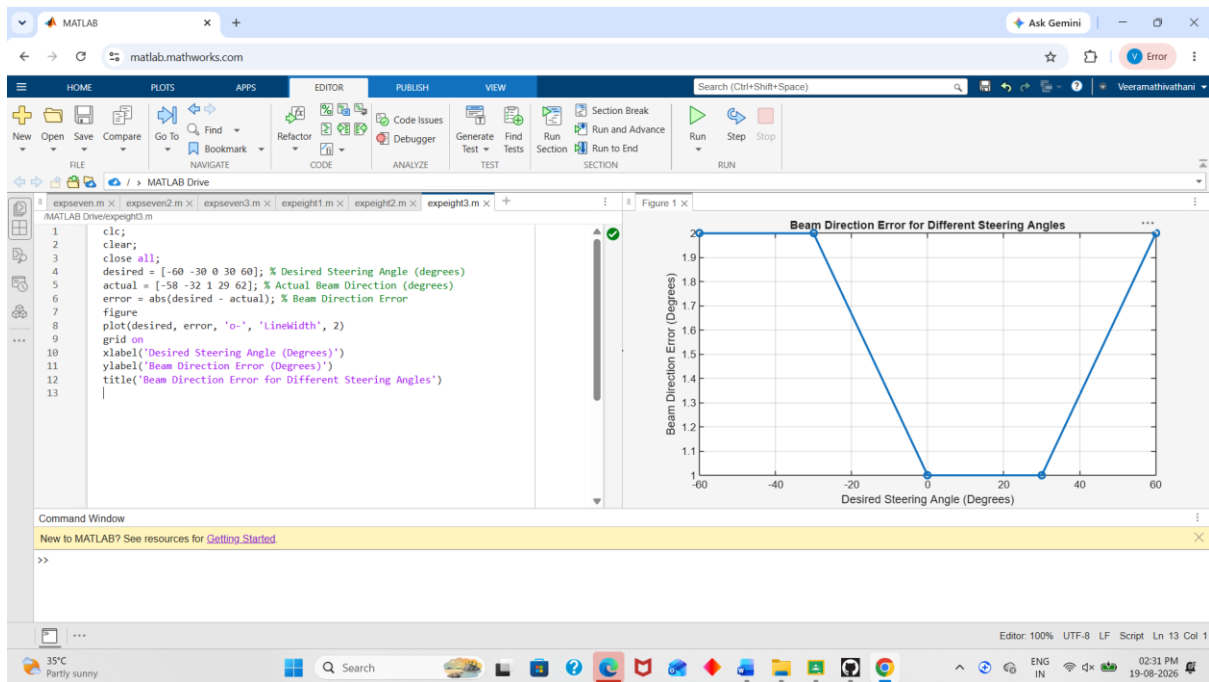
plot(desired, error, 'o-', 'LineWidth', 2)

grid on

xlabel('Desired Steering Angle (Degrees)')

ylabel('Beam Direction Error (Degrees)')

title('Beam Direction Error for Different Steering Angles')



Result:

The MATLAB analysis successfully demonstrated steering angle variation in a MIMO beamforming system. The results showed that beam steering effectively directs the transmitted energy toward different angles, while maintaining high array gain and minimizing beam direction error.